ITEC 116 – System Integration and Architecture

Group Activity 3: Module Integration (Finance & Accounting → Human Resources)

Group Number: 2

Sub-Team	Role	Members	Responsibilities
A	**HR System Engineers**	Charls Sampaga Kian Lhei Pagkaliwagan Stephanie April Romo	Define HR payroll output schema (Employee_ID, Base_Salary, Tax_Deductions, Net_Pay).
B	**Finance & Accounting Engineers**	MharGranado Messiahlyn Mae Mabilangan	Define Finance General Ledger input schema (Journal_Entry_ID, Account_Code, Debit_Amount, Credit_Amount).
C	**Integration & Middleware Architects**	Curlh Joseph Jordan Golez Laurence Barcarse	Map HR fields to Finance Ledger entries; write business logic for ACID compliance and transactional rollback.
D	**Security, Compliance & QA Testers**	Mike Casaverde Lanz Michael Casino	Define security protocols (OAuth 2.0 / Token Auth), audit logging rules, and assemble the final report.

Section 1: HR & Finance Payload Snippets

HR System Payload (Sub-Team A)

REST JSON request payload emitted by the HR module after payroll approval:

```
{
  "payrollTransactionId": "PAY-2026-08-0001",
  "employeeId": "EMP-10452",
  "employeeName": "Barcarse, Laurence M.",
  "department": "Information Technology",
  "payPeriod": "2026-08-01 to 2026-08-15",
  "payDate": "2026-08-20",
  "grossPay": 45000.00,
  "taxDeduction": 6750.00,
  "otherDeductions": {
    "sss": 1350.00,
    "philhealth": 900.00,
    "pagibig": 200.00
  },
  "totalDeductions": 9200.00,
  "netPay": 35800.00,
  "paymentMethod": "Bank Transfer",
  "status": "Approved"
}
```

Finance Ledger Payload (Sub-Team B)

Target JSON payload consumed by the Accounting system to post a balanced double-entry journal:

```
{
  "journalId": "JE-2026-08-00145",
  "referenceId": "PAY-2026-08-0001",
  "transactionDate": "2026-08-20",
  "entries": [
    { "lineNo": 1, "accountCode": "5001", "accountName": "Payroll Expense",
      "debitAmount": 45000.00, "creditAmount": 0.00 },
    { "lineNo": 2, "accountCode": "2010", "accountName": "Tax Liability Payable",
      "debitAmount": 0.00, "creditAmount": 9200.00 },
    { "lineNo": 3, "accountCode": "1010", "accountName": "Cash/Bank - Disbursement Account",
      "debitAmount": 0.00, "creditAmount": 35800.00 }
  ],
  "totalDebit": 45000.00,
  "totalCredit": 45000.00,
  "totalDisbursement": 35800.00,
  "balanceCheck": "Debits = Credits",
  "status": "Posted"
}
```

Equivalent SOAP/XML variant (for legacy ERP posting):

```
<?xml version="1.0" encoding="UTF-8"?>
<soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/">
  <soap:Header/>
  <soap:Body>
    <JournalEntry id="JE-2026-08-00145" reference="PAY-2026-08-0001">
      <TransactionDate>2026-08-20</TransactionDate>
      <Line accountCode="5001" accountName="Payroll Expense">
        <Debit>45000.00</Debit><Credit>0.00</Credit>
      </Line>
      <Line accountCode="2010" accountName="Tax Liability Payable">
        <Debit>0.00</Debit><Credit>9200.00</Credit>
      </Line>
      <Line accountCode="1010" accountName="Cash/Bank - Disbursement Account">
        <Debit>0.00</Debit><Credit>35800.00</Credit>
      </Line>
      <TotalDisbursement>35800.00</TotalDisbursement>
      <Status>Posted</Status>
    </JournalEntry>
  </soap:Body>
</soap:Envelope>
```

Section 2: Integration Mapping Table - Task 2: Cross-Module Transformation & Transaction Logic

Data Field Mapping & Transformation Matrix (Sub-Team C)

Source Field (HR)	Target Field (Finance)	Data Type / Transformation	Accounting / Business Logic Rule
employeeId	referenceId	String	Pass through as foreign key link between payroll and journal entry
payrollTransactionId	referenceId (prefix)	String	Used to generate journalId as JE-{yyyy}-{mm}-{seq}
payDate	transactionDate	Date (YYYY-MM-DD)	Direct cast; must match the GL posting date
grossPay	Debit_Amount (Line 1)	Currency / Float	Debits the Payroll Expense Account (Code 5001)
taxDeduction	Credit_Amount (Line 2)	Currency / Float	Credits the Tax Liability Account (Code 2010)
netPay	Credit_Amount (Line 3)	Currency / Float	Credits the Cash Disbursement Account (Code 1010)
grossPay - netPay	Credit_Amount validation	Calculated	Must equal taxDeduction + otherDeductions total
totalDebit	—	Calculated	Sum of all debit lines; must equal totalCredit (Debits = Credits rule)

Transactional Resiliency Check — 2-Phase Commit / ACID Rollback

If the network drops after HR marks payroll as "Approved" but before Finance confirms the journal entry is "Posted," the transaction must not be left half-applied. The integration layer prevents this using a two-phase commit (2PC) pattern:

1. **Prepare phase:** HR sends the payroll payload to the middleware, which stages it and returns a pending transaction ID without finalizing anything in either system.

2. **Vote phase:** Finance validates the staged journal entry (checks debits = credits, valid account codes) and returns a

ready-to-commit or abort vote.

3. **Commit phase:** Only if both HR and Finance vote "ready" does the middleware issue a commit to both systems atomically; otherwise it issues a rollback to both, reverting HR's payroll status back to Pending Disbursement and discarding the unposted journal entry.

4. **Timeout / recovery:** If the network drops mid-transaction, the transaction is flagged in-doubt. A recovery service polls both systems on reconnect and either completes the commit (if both sides logged the same transaction ID as prepared) or forces a full rollback — ensuring no orphaned payroll disbursement exists without its matching GL entry (ACID atomicity and consistency).

Section 3: Transactional & Security Architecture

Authentication (Sub-Team D)

OAuth 2.0 Client Credentials flow (system-to-system, no end user involved). The HR system authenticates against an internal Identity Provider to obtain a short-lived Bearer access token (e.g., 15-minute expiry), presented in the Authorization: Bearer <token> header on every call to the Finance API. Communication is additionally secured with mutual TLS (mTLS) so both HR and Finance services verify each other's certificates, preventing spoofed payroll submissions.

Audit & Compliance Logging Format

Structured JSON, written to an immutable/append-only log store, aligned with data-privacy requirements such as the Philippine Data Privacy Act (RA 10173):

{

"logId": "AUD-2026-08-0001", "timestamp": "2026-08-20T09:15:32Z",

"actorSystem": "HR-Payroll-Service", "action": "PAYROLL_DISBURSEMENT_POSTED",

"referenceId": "PAY-2026-08-0001", "journalId": "JE-2026-08-00145",

"authMethod": "OAuth2-ClientCredentials", "tokenSubject": "svc-hr-payroll", "ipSource": "10.12.4.51",

"result": "SUCCESS",

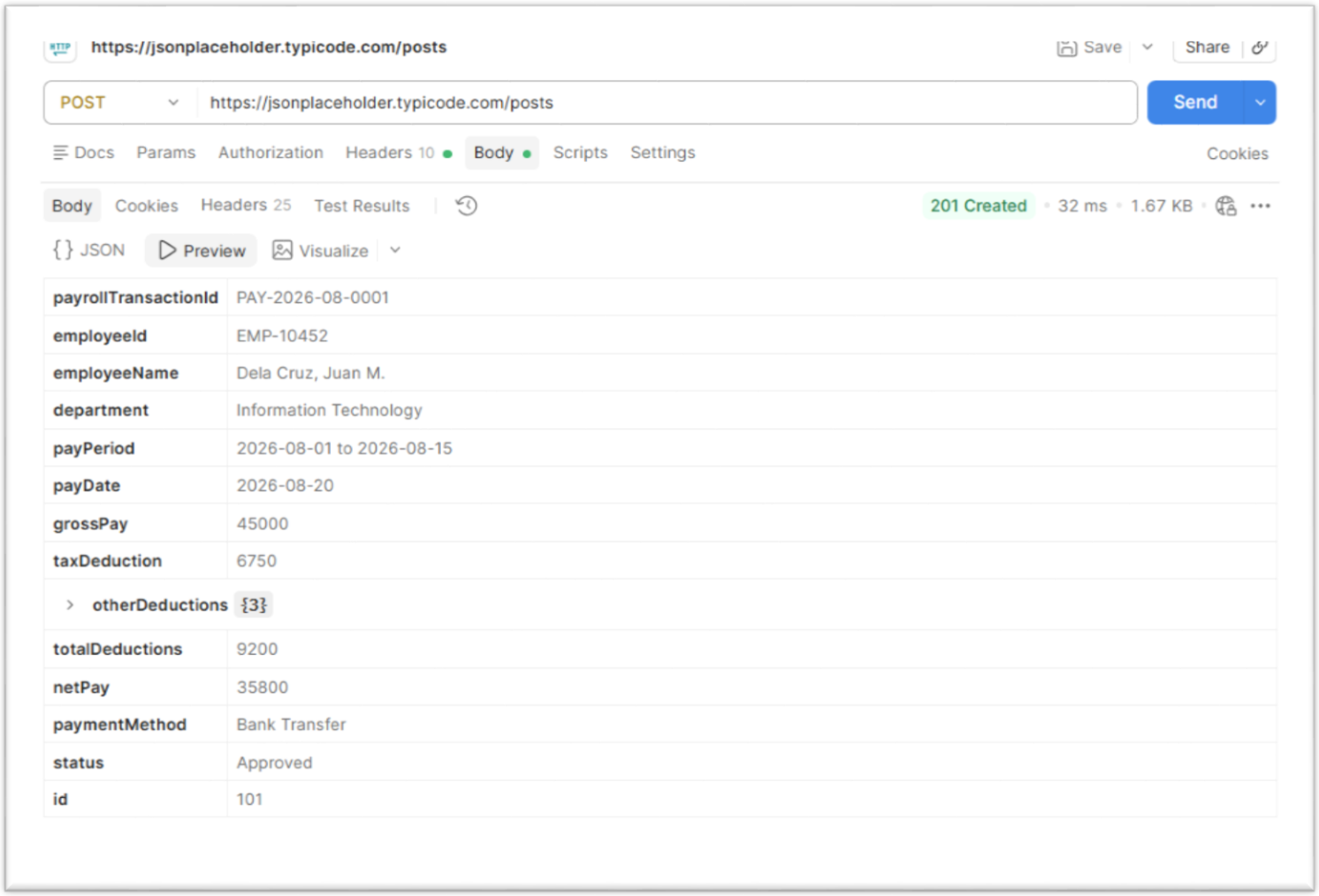
"sensitiveFieldsMasked": ["employeeName", "bankAccountNo"]

}

Key Policies

- Mask/tokenize PII (employee name, bank account number) in all log entries.
- Retain audit logs for a fixed compliance period (e.g., 5 years for financial records).
- Restrict log read-access to the Compliance/QA role only (principle of least privilege).
- Log both successful and failed/aborted transactions — critical for reconstructing the rollback scenario described in Task 2.

Postman Screenshot:



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Evaluation Rubric (50 Points Total)

- **HR & Finance Payload Alignment (15 pts):** JSON/XML schemas accurately capture payroll and financial journal entries.
- **Transformation & Accounting Logic (15 pts):** Field mapping correctly reflects accounting rules (Debits = Credits) and data casting.
- **Transactional Resiliency & Security (10 pts):** Clear explanation of failure rollbacks, API authentication, and security protocols.
- **Large Group Coordination (10 pts):** All 8–9 members are explicitly assigned tasks with seamless integration across sections.

Names of Teams:

A. 1. Charls Sampaga
2. Kian Lhei Pagkaliwagan
3. _____

B. 1. Mhar Granado
2. Messiahlyn Mae Mabilangan
3. _____

C. 1. Laurence Barcarse
2. Curlh Joseph Jordan Golez
3. Stephanie April Romo
4. _____

D. 1. Mike Casaverde
2. Lanz Michael Casino
3. _____